

Strong form

$$\begin{aligned}
(0.1) \quad & \check{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla (\phi_f^{-1/2} \check{q}_f) = 0 \\
& \mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} (\check{q}_f - \phi_f^{1/2} q) = 0 \\
& \nabla q - \nabla \cdot \hat{\sigma}(\mathbf{v}_s) = -(1 - \phi_f) \rho_r \underline{\mathbf{g}} \\
& \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} (\check{q}_f - \phi_f^{1/2} q) = 0
\end{aligned}$$

List SI units

q = pascal.

v = m/s

Non dimensionalization choice

- $\check{\mathbf{v}} = \frac{\mathbf{v}_r}{k_0 \rho_r g_y / \mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$

The resulting non dimensionalized form is therefore

$$\begin{aligned}
(0.2) \quad & \check{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \check{q}_f) = 0 \\
& \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \check{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0 \\
& \nabla \check{q} - \nabla \cdot \hat{\sigma}(\check{\mathbf{v}}_s) = -(1 - \phi_f) \rho_r \hat{\underline{\mathbf{g}}} \\
& \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0
\end{aligned}$$

resulting weak form